A greedy sequence that avoids sums of three consecutive terms

A proof of the first conjecture in OEIS A394576

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Abstract

The sequence A394576 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Hoang Xuan Thanh, starts with $a(1) = 1$ and then always takes the smallest integer larger than the previous term that is not a sum $a(k) + a(k+1) + a(k+2)$ of three consecutive earlier terms. Its first conjecture is the bound $a(n) \le \lfloor (4n-1)/3 \rfloor$. We prove the two-sided bound $\lfloor (4n-3)/3 \rfloor \le a(n) \le \lfloor (4n-1)/3 \rfloor$ for every $n \ge 1$. Since the two bounds differ by at most one, $a(n)$ is always within 1 of the upper bound, and $a(3k) = 4k-1$ exactly. The proof constructs the sequence in blocks of three: each interval $\{4k, 4k+1, 4k+2, 4k+3\}$ contains exactly one forbidden sum, and the other three numbers are the next three terms. We then prove that this construction satisfies the greedy rule, and that the rule has only one solution. The results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [3] is a public database of integer sequences. Many entries record *conjectures*: patterns observed in the data that the contributor could not prove. This paper concerns entry A394576 [1], contributed by Hoang Xuan Thanh in 2026, which reads in part:

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a(1) = 1; a(n+1) is the smallest positive integer greater than a(n) that
avoids violating the condition a(k) + a(k+1) + a(k+2) is not a term.

Conjecture 1:  a(n) <= floor((4*n-1)/3).
```

— OEIS A394576, Hoang Xuan Thanh, May 2026

In other words, the sequence is built one term at a time, and the rule is that no sum of three consecutive terms may itself be a term. When $a(n{+}1)$ is chosen, the sums $a(k){+}a(k{+}1){+}a(k{+}2)$ with $k+2 \le n$ are already known, and $a(n+1)$ must avoid them. A sum that involves the new term is larger than every term chosen so far, so it can only be avoided by later terms. Precisely, the rule is: $a(1) = 1$, and for $n \ge 1$,

$$a(n+1) = \min\{t > a(n) :\ t \ne a(k) + a(k+1) + a(k+2) \text{ for all } k \ge 1 \text{ with } k+2 \le n\}. \quad (1)$$

The first terms are 1, 2, 3, 4, 5, 7, 8, 10, 11, 13, 14, 15, 17, 18, 19, 21, 22, 23, 24, 26, ... For instance, $a(6) = 7$ because $a(1) + a(2) + a(3) = 6$ is forbidden.

1.2 The question

The first conjecture of the entry says that $a(n) \le \lfloor (4n-1)/3 \rfloor$, where $\lfloor y \rfloor$ denotes the largest integer not exceeding y. The table compares the first 32 terms with this bound and with the lower bound $\lfloor (4n-3)/3 \rfloor$ proved below. The indices in bold are the multiples of 3, where the two bounds coincide.

n	1	2	**3**	4	5	**6**	7	8	**9**	10	11	**12**	13	14	**15**	16
$\lfloor \frac{4n-3}{3} \rfloor$	0	1	**3**	4	5	**7**	8	9	**11**	12	13	**15**	16	17	**19**	20
$a(n)$	1	2	**3**	4	5	**7**	8	10	**11**	13	14	**15**	17	18	**19**	21
$\lfloor \frac{4n-1}{3} \rfloor$	1	2	**3**	5	6	**7**	9	10	**11**	13	14	**15**	17	18	**19**	21

n	17	**18**	19	20	**21**	22	23	**24**	25	26	**27**	28	29	**30**	31	32
$\lfloor \frac{4n-3}{3} \rfloor$	21	**23**	24	25	**27**	28	29	**31**	32	33	**35**	36	37	**39**	40	41
$a(n)$	22	**23**	24	26	**27**	28	30	**31**	32	33	**35**	36	37	**39**	40	41
$\lfloor \frac{4n-1}{3} \rfloor$	22	**23**	25	26	**27**	29	30	**31**	33	34	**35**	37	38	**39**	41	42

The upper bound $\lfloor (4n-1)/3 \rfloor$ is the n-th positive integer not divisible by 4 (OEIS A042968): the sequence follows the numbers $1, 2, 3, 5, 6, 7, 9, \ldots$ closely, but occasionally falls one step behind.

1.3 What this paper proves

Main result (Theorem 3.1). *For every $n \ge 1$,*

$$\left\lfloor \frac{4n-3}{3} \right\rfloor \le a(n) \le \left\lfloor \frac{4n-1}{3} \right\rfloor.$$

The upper bound is Conjecture 1 of the entry. The two bounds differ by at most 1 and agree when n is a multiple of 3, so $a(3k) = 4k - 1$ for every $k \ge 1$ (Corollary 3.2).

1.4 What was known

The entry records further conjectures (Conjectures 2 and 3) describing in detail where $a(n)$ equals the upper bound and where it falls one short; they are not proved here. Thanh has deposited software that computes the sequence and its generalizations [2]. When the entry was checked on 24 October 2026, the bound was still labelled as a conjecture, and we are not aware of an earlier proof.

1.5 Why it is interesting

Greedy sequences that avoid a forbidden pattern are a classical source of hard problems: the rule is simple, but the global behaviour of the sequence is often irregular and difficult to predict. Here the behaviour turns out to be very regular. The forbidden sums fall exactly one into each block $\{4k, 4k+1, 4k+2, 4k+3\}$, so the sequence takes three out of every four integers, and it has density $3/4$ in the positive integers. The proof shows how a greedy definition can be replaced by an explicit construction, and how to prove that the two agree.

1.6 How to read this paper

Section 2 fixes the notation. Section 3 states the results. Section 4 gives the block construction and proves that it is the greedy sequence. Section 5 describes the computer checks and the formal proof, and Section 6 discusses the remaining conjectures.

2 Background

2.1 Notation

For a real number y, $\lfloor y \rfloor$ is the largest integer not exceeding y. We write

$$L(n) = \Big\lfloor \frac{4n-3}{3} \Big\rfloor, \qquad U(n) = \Big\lfloor \frac{4n-1}{3} \Big\rfloor \qquad (n \geq 1).$$

Writing $n = 3k + r$ with $r \in \{1, 2, 3\}$, one finds

n	$3k+1$	$3k+2$	$3k+3$
$L(n)$	$4k$	$4k+1$	$4k+3$
$U(n)$	$4k+1$	$4k+2$	$4k+3$

(2)

for every $k \geq 0$. In particular $0 \leq U(n) - L(n) \leq 1$, with equality $L(n) = U(n)$ when $3 \mid n$.

2.2 Greedy definitions

A sequence defined by a rule of the form (1), in which each new term is the least admissible integer, is called *greedy*. Two sequences that satisfy (1) and agree up to index n have the same forbidden sums at that stage, hence the same least admissible next term; by induction, (1) has exactly one solution. The difficulty in proving properties of greedy sequences is that the rule is local, while statements such as a bound for $a(n)$ are global. The proof below replaces the rule by an explicit construction.

3 Results

Theorem 3.1. *The rule* (1) *has exactly one solution, and for every* $n \geq 1$,

$$L(n) \leq a(n) \leq U(n).$$

Moreover, the terms of the sequence and the sums $a(k) + a(k+1) + a(k+2)$, $k \geq 1$, *are disjoint sets whose union is the set of all positive integers.*

Corollary 3.2. *For every $n \geq 1$, $U(n) - a(n) \in \{0, 1\}$; in particular Conjecture 1 of the entry holds. For every $k \geq 1$, $a(3k) = 4k - 1$.*

Proof. This follows from Theorem 3.1 and the table (2). □

4 Proofs

The idea in brief. We build a sequence A in blocks of three. The first block is $1, 2, 3$. At stage $k \geq 1$ the sum $b(k) = A(k) + A(k+1) + A(k+2)$ is already known, and the next three terms are the elements of $\{4k, 4k+1, 4k+2, 4k+3\}$ other than $b(k)$. The bounds $L \leq A \leq U$ show that $b(k)$ always lies in this block, so the construction never fails. Finally, every integer that the construction skips is one of the sums $b(k)$, which shows that A obeys the greedy rule.

4.1 The block construction

Put $A(1) = 1$, $A(2) = 2$, $A(3) = 3$. Suppose $A(1), \ldots, A(3k)$ have been defined, with $k \geq 1$. Since $k + 2 \leq 3k$, the number

$$b(k) = A(k) + A(k+1) + A(k+2) \tag{3}$$

is known. Let $A(3k+1) < A(3k+2) < A(3k+3)$ be the elements of $\{4k, 4k+1, 4k+2, 4k+3\} \setminus \{b(k)\}$, listed in increasing order (if $b(k)$ is not in the block, take its three smallest elements; Lemma 4.2 shows that this case never occurs). For example, $b(1) = 6$ gives the block $4, 5, 7$, and then $b(2) = 2 + 3 + 4 = 9$ gives $8, 10, 11$.

Lemma 4.1. *For every $k \geq 1$, $L(k)+L(k+1)+L(k+2) = 4k$ and $U(k)+U(k+1)+U(k+2) = 4k+2$.*

Proof. The three numbers $4k - 3$, $4k + 1$, $4k + 5$ have the three different remainders $0, 1, 2$ modulo 3, in some order. Subtracting these remainders and dividing by 3 gives $L(k) + L(k+1) + L(k+2) = \frac{(12k+3)-3}{3} = 4k$. The same argument with $4k - 1$, $4k + 3$, $4k + 7$ gives $U(k) + U(k+1) + U(k+2) = \frac{(12k+9)-3}{3} = 4k + 2$. □

Lemma 4.2. *For every $n \geq 1$, $L(n) \leq A(n) \leq U(n)$, and for every $k \geq 1$, $4k \leq b(k) \leq 4k+2$.*

Proof. The bounds hold for $n = 1, 2, 3$ by (2). Suppose they hold up to $n = 3k$. Since $k + 2 \leq 3k$, Lemma 4.1 gives $4k \leq b(k) \leq 4k + 2$. Write $b(k) = 4k + s$ with $s \in \{0, 1, 2\}$. Removing $4k + s$ from $\{4k, \ldots, 4k + 3\}$ leaves a smallest element in $\{4k, 4k+1\}$, a middle element in $\{4k+1, 4k+2\}$, and the largest element $4k+3$. By (2) these are exactly the ranges $[L(n), U(n)]$ for $n = 3k + 1, 3k + 2, 3k + 3$. This completes the induction. □

Lemma 4.3. *The sequence A is strictly increasing. Its terms and the numbers $b(k)$, $k \geq 1$, form two disjoint sets whose union is the set of positive integers.*

Proof. The first block is $\{1, 2, 3\}$. By Lemma 4.2, the block of stage k consists of the four integers $4k, \ldots, 4k+3$ except $b(k)$, listed in increasing order. These intervals cover the integers from 4 on without overlap, each contains exactly one $b(k)$, and different k give different intervals. □

Proof of Theorem 3.1. We show that A satisfies (1). Fix $n \geq 1$. First, $A(n+1) > A(n)$, and $A(n+1)$ is not equal to any $b(k)$, by Lemma 4.3; so $A(n+1)$ is admissible. Next, let t be an integer with $A(n) < t < A(n+1)$. By Lemma 4.3, $t = b(k)$ for some $k \geq 1$. Since all terms are positive, $A(k+2) < b(k) = t < A(n+1)$, and since A is increasing, $k+2 < n+1$, that is $k+2 \leq n$. So t is one of the sums forbidden at stage n, and $A(n+1)$ is the least admissible integer. Hence A satisfies (1). As explained in Section 2, (1) has only one solution, so $A = a$. The bounds now follow from Lemma 4.2, and the last statement from Lemma 4.3. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed the sequence directly from the greedy rule (1), for $n \leq 100{,}000$. This reproduces all 75 terms printed in the OEIS entry, and confirms the bounds of Theorem 3.1, the values $a(3k) = 4k - 1$, and that no term is a sum of three consecutive terms, throughout that range. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [8]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [6] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [7], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A394576.lean` of the repository, in the namespace `OEIS.A394576`. There the sequence is indexed from 0: `sequence n` is $a(n+1)$, and the rule is written out explicitly in the statement of `source_existence_uniqueness`. The construction and its properties are in `lean/Families/GreedyTriple.lean` (namespace `Families.GreedyTriple`). The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
The rule (1) has exactly one solution	`source_existence_uniqueness`
The block construction satisfies (1)	`canonical_source`
Theorem 3.1, the bounds	`canonical_bounds`, `source_implies_bound`
Theorem 3.1, the partition of the positive integers	`complementary_ranges`
Corollary 3.2	`bounded_error`, `exact_every_third`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.1 proves Conjecture 1 of the entry and a matching lower bound. It does not prove Conjectures 2 and 3, which describe exactly where $a(n) = U(n)$ and where $a(n) = U(n) - 1$. In the language of this paper, they describe the position $s = b(k) - 4k \in \{0, 1, 2\}$ of the forbidden sum in each block: by the proof of Lemma 4.2, the differences $U(n) - a(n)$ for $n = 3k + 1, 3k + 2, 3k + 3$ are $(0, 0, 0)$, $(1, 0, 0)$ or $(1, 1, 0)$ according as $s = 0$, 1 or 2. The words $(1, 0, 0)$ and $(1, 1, 0)$ are the building blocks X and Y of Conjecture 3.

The skipped numbers. By Theorem 3.1, the positive integers not in the sequence are exactly the sums $b(k) = a(k) + a(k + 1) + a(k + 2)$, $k \geq 1$: $6, 9, 12, 16, 20, \ldots$, one in each interval $\{4k, \ldots, 4k+3\}$. A finer description of the numbers $b(k) - 4k$ would give Conjectures 2 and 3.

Sums of more terms. The entry refers to the analogous sequence A394676, in which sums of four consecutive terms are forbidden. A block argument of the same kind would require an interval that contains exactly one forbidden sum, and it would be interesting to know whether such a structure exists there.

About this paper

How the work was done. This paper is one of 29 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [8]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [8], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and its conjectures are due to Hoang Xuan Thanh [1], who also deposited software that computes the sequence [2].

References

[1] H. X. Thanh, entry A394576 ($a(n+1)$ is the smallest integer greater than $a(n)$ avoiding sums of three consecutive terms), *The On-Line Encyclopedia of Integer Sequences*, May 2026. `https://oeis.org/A394576`. Text as of October 2026.

[2] H. X. Thanh, *Algorithm for OEIS A394576, A394676 and generalized forbidden sums*, software deposit, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22674326`.

[3] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[4] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[5] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[6] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[7] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `c44e0c8ee63c`.

[8] D. D. Goyal, *Proofs and corrections of OEIS conjectures, with Lean 4 formalizations*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.